

Take-Home Assignment – AI Knowledge Assistant

Overview

Build a simple **AI Knowledge Assistant** that allows users to register, log in, and interact with an AI assistant through a chat interface.

The application should demonstrate your full-stack development skills, including authentication, REST API development, database design, AI integration, and deployment.

For the AI integration, use **OpenRouter** with any **free model**.

Functional Requirements

Authentication

Implement a secure authentication system with:

- User Registration
- User Login
- User Logout
- JWT Authentication
- Secure password hashing

Only authenticated users should be able to access the application.

AI Chat

After logging in, users should be able to:

- Start a new conversation
 - Send prompts to the AI
 - View AI responses
 - Continue existing conversations
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Conversation History

Users should be able to:

- View all previous conversations
- Continue a conversation
- Delete a conversation

Each conversation should be stored in the database.

Technical Requirements

Frontend

- React (Vite or Next.js)
- TypeScript
- Responsive UI
- Proper error handling and loading states

Backend

- Node.js (Express or NestJS)
- TypeScript
- REST API
- JWT Authentication

Database

- PostgreSQL

AI Integration

- OpenRouter
 - Use any **Free Model**
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Deployment

Deploy your application using **free hosting services**.

Suggested platforms:

- **Frontend:** Vercel or Netlify

- **Backend:** Render, Railway, or Fly.io
 - **Database:** Supabase PostgreSQL or Railway PostgreSQL
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Submission

Please submit:

- GitHub Repository
 - Live Demo URL
 - README containing:
 - Setup instructions
 - Technologies used
 - How to run the project locally
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Evaluation Criteria

Your submission will be evaluated based on:

- Code Quality
 - Project Structure
 - UI/UX
 - Authentication & Security
 - AI Integration
 - Database Design
 - Error Handling
 - Overall Application Quality
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Submission Deadline

Please submit your completed assignment by **Monday, 27 July 2026**.

Notes

- Use **only free-tier services**.
- The AI integration **must use OpenRouter with a free model**.
- Store all secrets using **environment variables**.
- Ensure your code is clean, maintainable, and well-structured.
- Feel free to add additional features if you wish, but the core requirements should be completed first.
- During the follow-up interview, you should be able to explain your implementation, architecture, and technical decisions.